

Robot Movement Planning

Planning approach:

Free-space planning – implemented when the arm needs to reach a specific safe position.

Cartesian planning – employed when the end-effector moves in a controlled and straight line, either descending towards the battery or lifting vertically.

1. Movement sequence overview

Stage	Movement type	Purpose	Main check
1. Pre-grasp	Free-space	Move above the battery	No collision; correct tool orientation
2. Approach	Cartesian	Descend vertically	Suction cup aligned with battery
3. Grasp	Vacuum command	Create suction seal	Vacuum confirmed
4. Extraction	Cartesian	Lift battery vertically	Battery clears phone safely
5. Transfer and release	Free-space + release	Move to disposal area	Battery released and detached

2. Move to pre-grasp

Once directions/coordinates are received from the computer vision team, the robot moves into the first phase – pre-grasp. The purpose of this phase is to move above the battery; make sure alignment is central and steady for the cartesian approach in the next phase. Now in terms of offset, 10cm is what we have currently defined, however can be adjusted based on physical inspection and testing in the lab.

The movement type of this phase should use free-space planning. Unless we want it to, the robot doesn't have to follow a strict path to the coordinates above the battery at this stage, however, we must make sure that the environment surrounding the robot is clear and won't collide with anything in the space.

Additionally, we ideally have the end-effector in a set orientation that ensures that the suction to the battery is secure and importantly is facing the battery surface before initiating its descent down. This would remove any rotation outliers as well as unsecure grasping by the suction cup.

If not tested yet, the arm offset before descent should be tested to find the right middle ground to where it is moving efficiently and not travelling more than it needs to, is in a suitable position for the vertical extraction phase and has enough room for the suction cup to grasp.

3. Cartesian approach - straight-line descent

Once the robot is positioned above the battery and in alignment with the CV team's coordinates, the second phase of straight-line descent can be initiated with a cartesian planner. A cartesian planner would be important here as it would keep the end effector on a simple and straight path, descending to the battery and maintaining its orientation.

Important to note, this phase while descending would achieve better results and a more accurate suction to the battery if we implement a slower descent speed compared to the previous free-space movement, and moreover,

should stop its descent once proper contact is reached – this contact condition can be defined through physical testing.

To ensure that this phase goes as smoothly as possible, we may list down a few requirements:

- There should be a fixed location for each fixed phone
- Suction cup should make contact with the flat surface, central area of the battery for proper suction
- We should determine a constant end-effector orientation – however, may be useful to think of differing physical battery conditions and uneven surfaces, and how we can implement the end-effector to adjust to this
- We should define what our contact condition/condition for ending the descent should be to avoid any unnecessary force
- If for any reason the target pose is unreachable, the arm should be able to abort and return to its start state

4. Vacuum grasp and Planning Scene attachment

As contact has been made with the battery, the vacuum gripper should be activated. This phase can be overlooked, as the arm shouldn't take a successful command as a successful grasp – essentially, it shouldn't assume that since it made contact with the battery and that the vacuum gripper has been activated, that the battery is securely attached to the end-effector.

For this, we should have a sequence resembling something like:

1. Vacuum gripper activation once battery contact has been achieved
2. Important – confirmation of successful vacuum suction via a pressure sensor or signal
3. If no confirmation has been received (maybe during a certain amount of time we set), then stop, retry, or finally abort and return to start state
4. However, if confirmation is received, attach the battery collision object to the end-effector in the MoveIt Planning Scene
5. Finally, begin extraction once attachment state is confirmed

To note, by attaching the battery in the Planning Scene, this tells MoveIt that the robot actually has the battery and is moving with it. If this isn't set, the arm may move and avoid collisions, not accounting for the physical battery attached – could lead to collisions with the environment or surrounding objects.

5. Cartesian extraction - vertical lift

The next phase is the vertical lift once the suction grasp has been confirmed and the battery has been attached to the MoveIt Planning Scene. Again, cartesian planning should be implemented here as the battery should move in a direct upward direction, ensuring no collisions occur and that the battery remains attached to the end-effector. Similar to the descent, the speed in this phase should also be slow and controlled – moving upwards until the battery is completely free of the phone and the jig before beginning sideways transfer.

During this phase, we should monitor and take the following data:

- Grasp status/vacuum pressure
- If we have a clean battery extraction – monitoring whether the phone or jig has moved
- Arm movement speed and joint-torque feedback (if we can)

6. Transfer, release and return

After we have confirmed that the battery is clear of the phone, we return to free-space planning and begin the transfer, release and return phase – moving to a pre-defined drop-off/disposal zone. While this is happening, we still have the battery attached to the end-effector in the MoveIt Planning Scene – allowing it to make collision detection decisions accounting for the added battery.

Once the release pose has been reached, we can switch the vacuum suction off and the battery should be removed from the end-effector in the Planning Scene as we have now returned to the initial state with just the arm. Then, the robot can return to the home/ready pose for the next battery.

So, simply, the release sequence could look something like:

- Move to stable position above the disposal location
- Vacuum deactivation
- Battery release confirmation
- Detach the battery collision object
- Withdraw and return to the ready state

7. RIoT Lab testing

Some parts of the movement plan cannot be validated properly through RViz or Gazebo alone. The following should be done in the RIoT Lab – some with accordance with the other sub-teams.

7.1 Hand-eye calibration

Since the camera and robot must share an accurate spatial relationship, we should liaise with the CV team as to how the hand-eye calibration will be implemented to determine the transform – translating what the camera can see into coordinates that the arm can map and move to.

We should test:

- How accurately the arm can reach the camera's identified physical point, and if the end-effector can accurately land in the center of the battery
- Note whether there are any position or orientation errors that remain after calibration
- If there are any areas in the camera's vision that the transform is more consistent than others
- If recalibration is required for each session – for example, if the hardware/base is moved, will the same transform matrix still be accurate?

7.2 Real-world obstacle avoidance and integration testing

This is an important physical test that we cannot accurately test just with the RViz or Gazebo, as the real environment would be different – the arm itself, the table, phone fixture, phone position, end-effector, cables etc. All these aspects may have differences that cannot be accurately measured in the simulation environment alone.

We should confirm that:

- The Planning Scene matches as accurately as possible with the real work environment
- Arm aborting if an obstacle is detected blocking in the path
- We have a constant clearance test for pre-grasp and battery transfer path phases

7.3 Vacuum-gripper mitigation testing

The vacuum-gripper is something that can cause many unexpected outcomes. Therefore, the lab tests should be utilized to understand how we can respond to events such as incomplete seals, variation of surfaces and the effect this has on the end-effector and vacuum loss.

Risk	Mitigation to test	Expected response
No seal	Pressure threshold and timeout	Abort before lifting
Poor alignment	Compliant mount and controlled re-approach	Withdraw, realign and retry
Vacuum loss during lift	Continuous pressure monitoring	Stop and move to a safe state
Battery does not release	Force/torque limit	Stop extraction and request intervention
Battery rotates	Cup placement or dual-cup evaluation	Revise end-effector setup

8. To Sum Up

So to conclude, the robot movement plan would employ free-space planning when in the pre-grasp phase and moving through space to the target location above the battery and then implementing cartesian planning for the controlled descent phase. These stages would be organized into one task via the MTC, and MoveIt would handle the trajectories that involve the collision-awareness and whether the vacuum system is able to secure the battery or not.

Elements such as approach speed and distance, extraction limits, inspection of vacuum-gripper in action and calibration accuracy can all be confirmed through RIoT Lab testing.